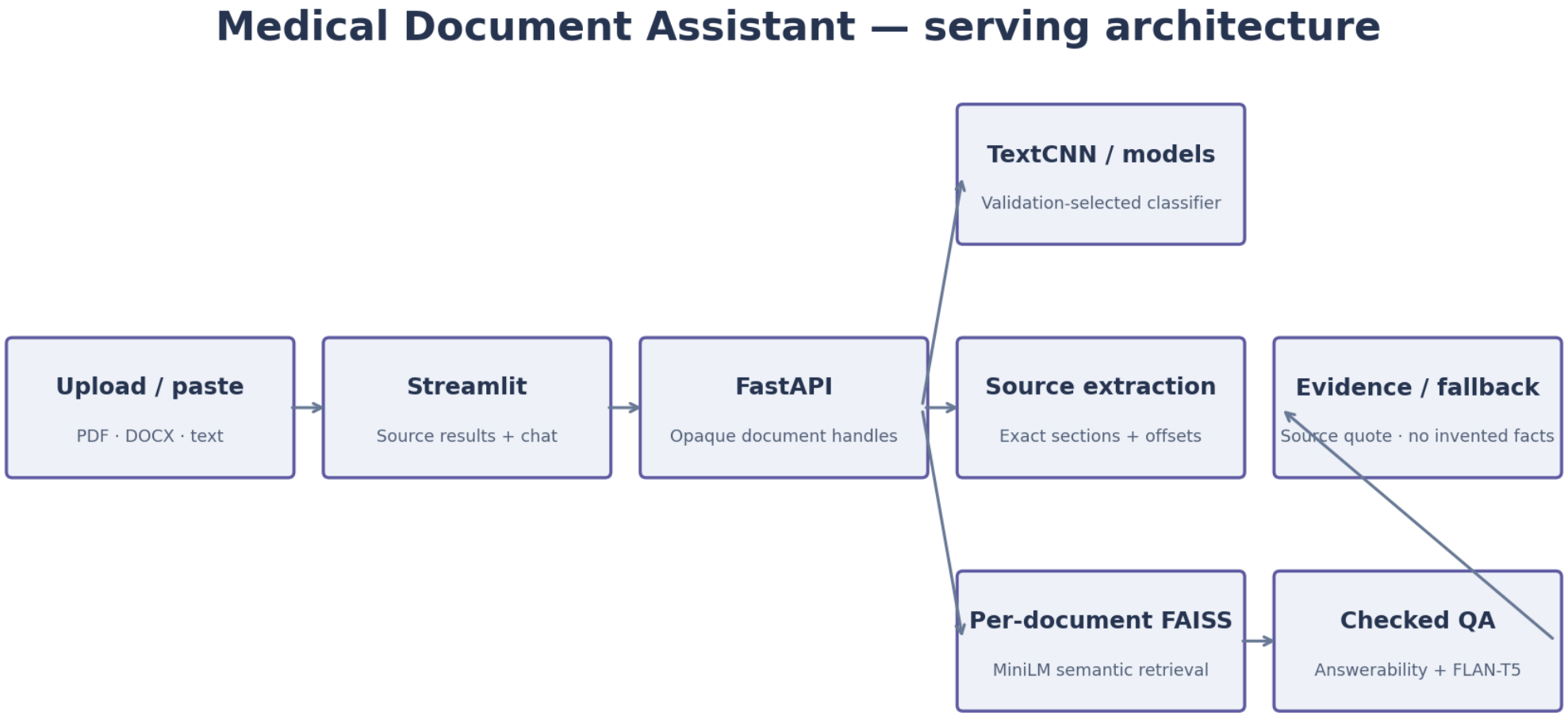
Medical Document Assistant

AI Engineering capstone · Python → ML → Deep Learning → NLP → Web app

English medicine-label classification and source-quoted document QA

Informational only — not medical advice

Architecture and application flow



Real data and fair evaluation

3,572 raw openFDA records → 2,471 cleaned English labels

Six explicit EPC-derived course categories

Frozen split: 1,762 train / 367 validation / 342 test

100 ingredient-aware components; source and split hashes recorded

Classifier comparison

Logistic Regression: accuracy 97.66% · macro F1 0.9799

Linear SVM: accuracy 95.61% · macro F1 0.9622

Random Forest: accuracy 97.08% · macro F1 0.9751

PyTorch TextCNN: accuracy 98.54% · macro F1 0.9876

Fine-tuned BERT-mini: accuracy 95.32% · macro F1 0.9597

Grounded QA and honest testing

MiniLM embeddings · per-document FAISS · local FLAN-T5

Answers are source quotations with exact character offsets

Missing evidence: 'I cannot find that in the document.'

20 regression cases / 3 evaluation leaflets: 11 answerable, 9 missing

Live document demonstration

Load the real FDA example or upload a text-based PDF

Inspect the predicted category and dosage/warning source sections

Ask: 'How long can this product take for full effect?'

Ask an absent fact, then clear the document

Verification and deployment

Real API/browser integration: 14 checks passed

Upload, classification, extraction, citations, fallbacks, and clearing tested

Docker image + Compose + Hugging Face Docker Space package

Temporary review URL and persistent-hosting handoff documented

Conclusion and next improvements

Complete educational data-to-application pipeline

Strongest recorded classifier: TextCNN · 98.54% accuracy / 0.9876 macro F1

Explicit limits: source-derived labels, small QA benchmark, English text

Extensions: evaluated Arabic models, OCR, broader QA cases, persistent hosting